

An Emulator for Fine-Tuning Large Language Models using Small Language Models

Eric Mitchell, Rafael Rafailov, Archit Sharma,
Chelsea Finn, Christopher D. Manning
Stanford University
eric.mitchell@cs.stanford.edu

Abstract

Widely used language models (LMs) are typically built by scaling up a two-stage training pipeline: a pre-training stage that uses a very large, diverse dataset of text and a fine-tuning (sometimes, ‘alignment’) stage that uses targeted examples or other specifications of desired behaviors. While it has been hypothesized that knowledge and skills come from pre-training, and fine-tuning mostly filters this knowledge and skillset, this intuition has not been extensively tested. To aid in doing so, we introduce a novel technique for decoupling the knowledge and skills gained in these two stages, enabling a direct answer to the question, *What would happen if we combined the knowledge learned by a large model during pre-training with the knowledge learned by a small model during fine-tuning (or vice versa)?* Using an RL-based framework derived from recent developments in learning from human preferences, we introduce *emulated fine-tuning (EFT)*, a principled and practical method for sampling from a distribution that approximates (or ‘emulates’) the result of pre-training and fine-tuning at different scales. Our experiments with EFT show that scaling up fine-tuning tends to improve helpfulness, while scaling up pre-training tends to improve factuality. Beyond decoupling scale, we show that EFT enables test-time adjustment of competing behavioral traits like helpfulness and harmlessness without additional training. Finally, a special case of emulated fine-tuning, which we call *LM up-scaling*, avoids resource-intensive fine-tuning of large pre-trained models by ensembling them with small fine-tuned models, essentially emulating the result of fine-tuning the large pre-trained model. Up-scaling consistently improves helpfulness and factuality of instruction-following models in the Llama, Llama-2, and Falcon families, without additional hyperparameters or training.

1 Introduction

Widely used instruction-following large language models (LLMs) typically follow a two-stage training procedure, with a stage of unsupervised pre-training on a large, diverse dataset followed by supervised fine-tuning on a much smaller, carefully curated dataset (Raffel et al., 2020; Chung et al., 2022). While both stages are important in producing models that possess broad world knowledge and perform a given task reliably, identifying exactly what capabilities emerge in which stage and at what scale is difficult (Wei et al., 2022; Schaeffer et al., 2023). For example, pre-trained models typically require careful prompting in order to perform a task; after fine-tuning for instruction following, they typically do not. Evaluation of the extent to which the core capability of ‘instruction following’ is learned during pre-training vs. during fine-tuning is thus seriously complicated by the choice of this prompt. To enable more direct attribution of capabilities to a stage of training, we introduce a principled technique for emulating the result of combining the capabilities gained from pre-training and fine-tuning at different model scales; see Figure 1. This technique, which we call emulated fine-tuning (EFT), enables: a) direct study of the capabilities that change as only one stage is scaled up or down; b) the practical benefit of approximating the result of fine-tuning a large model without the associated computational expense; and c) the ability to modify the fine-tuning objective (e.g., the tradeoff between helpfulness and harmlessness) at test time, without additional training.

Emulated fine-tuning is based on a simple factorization of the logits of a fine-tuned language model into a) the base log probabilities of a pre-trained base model and b) the ‘behavior delta’, or the difference between the log probabilities of a base model and fine-tuned model. This delta is a compact representation of the behavior change learned in fine-tuning and can be justified through either a reinforcement learning (Rafailov et al., 2023) or Bayesian inference (Korbak et al., 2022) framework. EFT thus emulates the result of pre-training at one scale and fine-tuning at another by adding base log probabilities computed by a model at

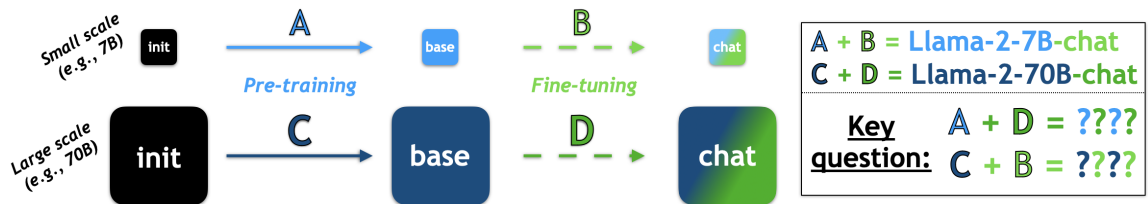


Figure 1: **Emulated fine-tuning (EFT)** enables a principled answer to the question of *what happens when we combine what is learned from pre-training a model of one size with what is learned from fine-tuning a model of a different size?* Conventional models combine the learnings of pre-training and fine-tuning at the same size ($A + B$, $C + D$). In contrast, EFT enables choosing these independently, allowing a principled approach to evaluating the result of $A + D$ and $C + B$.

one size and the behavior delta computed by models of a different size. For example, using models from the Llama-2 family, we can emulate the result of pre-training at 70B scale and fine-tuning at 7B scale by performing the log probability algebra **Llama-2-base 70B + (Llama-2-chat 7B - Llama-2-base 7B)**, where the first term is the base log probabilities and the term in parentheses is the behavioral delta. See Figure 2 for a concrete example of this emulation.

Using emulated fine-tuning, we analyze the results of pre-training and fine-tuning at various scales for multiple model families and datasets. Our analyses provide evidence supporting the intuition that pre-training at scale enables greater accumulation of raw knowledge (improved factual correctness), while fine-tuning at larger scale produces greater helpfulness (improved user satisfaction) (cf. [Gudibande et al., 2023](#)). Beyond this scientific finding, we also find that EFT enables boosting the performance of small fine-tuned models by a process we call *up-scaling*, essentially ensembling the small fine-tuned model with a larger pre-trained model, without any fine-tuning or modifications to either model. Our experiments show that in scenarios where fine-tuning a small language model is viable (e.g., Falcon-7B) but fine-tuning a larger language model is not due to resource constraints (e.g., Falcon-180B), up-scaling enables capturing much of the benefits of fine-tuning the larger model, without performing any model fine-tuning. Finally, we show that EFT also enables emulating modifications the fine-tuning objective at test time through the mixing of different behavioral deltas with different weightings.

In summary, our primary contributions are a) the emulated fine-tuning framework; b) clear experimental justification for the claim that scaling pre-training leads to improved factual knowledge while scaling fine-tuning leads to improved task adherence; and c) the technique of model *up-scaling*, which enables a small fine-tuned model and large base model to approximate the compute-intensive result of fine-tuning a large base model.

2 Related Work

The benefits of unsupervised pre-training in neural networks was first identified in deep belief networks ([Hinton et al., 2006](#)) and stacked autoencoders ([Bengio et al., 2007](#)), with early analyses noting persistent effects of pre-training even when fine-tuning data is not limited ([Erhan et al., 2010](#)). In natural language processing, pre-trained representations of individual words ([Mikolov et al., 2013](#); [Pennington et al., 2014](#)) or entire passages ([Devlin et al., 2019](#); [Peters et al., 2018](#)) demonstrated the ability for task-agnostic pre-training to learn representations useful for a wide variety of downstream linguistic tasks such as question-answering, natural language inference, and translation ([Devlin et al., 2019](#); [Raffel et al., 2020](#)). The transformer architecture ([Vaswani et al., 2017](#)) enabled more efficient pre-training on large datasets, which proved to inject significant amounts of precise factual world knowledge into pre-trained LMs ([Petrone et al., 2019](#)) that can be redirected to downstream tasks through fine-tuning ([Roberts et al., 2020](#)). Most recently, various works have shown that language models pre-trained with unsupervised generative modeling can be fine-tuned to engage in general-purpose dialogue, producing a model that can perform a variety of complex tasks specified in natural language ([Thoppilan et al., 2022](#); [Ouyang et al., 2022](#); [Bai et al., 2022](#); [Bubeck et al., 2023](#); [Touvron et al., 2023b](#)). Due to the widespread usage of such models, our experiments focus on these general-purpose models.

Increasing model scale has proven a key aspect of increasing the benefits of pre-training to fluency, world knowledge, reasoning ability, and a variety of other properties ([Brown et al., 2020](#); [Kaplan et al., 2020](#);

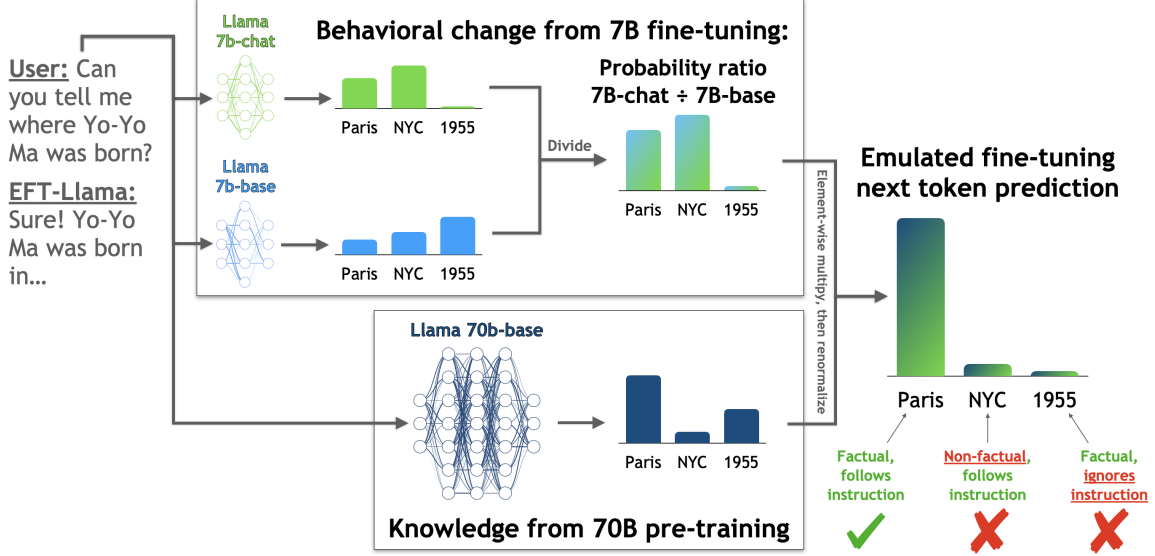


Figure 2: **Emulated fine-tuning combines knowledge from pre-training and fine-tuning at different scales.** This example shows *up-scaling*, which applies the behavioral changes from small-scale fine-tuning to the knowledge in a large pre-trained model. The small fine-tuned model (green) understands the user’s query asks about Yo-Yo Ma’s place of birth, not year, does not know the correct city. The small pre-trained model (light blue) does not understand the user’s query or have reliable knowledge, assigning high probability to the (correct) year of birth of Yo-Yo Ma and both possible places of birth. Their ratio represents the behavior of following user intent (responding only with locations). Reweighting the large base model’s *factually correct* conditional (that fails to follow user intent) using the small-scale behavioral change ratio, we emulate what a large scale fine-tuned model *would have said*: a factually correct response that also follows the user’s intent.

Touvron et al., 2023a). Other work leverages this capability differential to improve language model sampling through ‘contrastive decoding’, subtracting the log probabilities of a small language model (scaled by a small constant hyperparameter) from the log probabilities of a large language model (Li et al., 2023). Our work differs by interpreting this log probability difference as a log-importance weight, using it to re-weight the log probabilities of another model and eliminating the need for the added scaling hyperparameter. Relatedly, Gao et al. (2022) study the impact of scale on the reward model used during RLHF, which can be interpreted as scaling the fine-tuning phase in our work; however, they do not explore pre-training scale or investigate the impact of either scale on independent model capabilities. In concurrent work, Deng & Raffel (2023) train a reward model that reweights a base model’s conditional distributions during sampling. Our work differs in that EFT does not require training a new reward model, has a principled basis in reinforcement learning, and scales more efficiently with the vocabulary size, due to the parameterization of the reward as a ratio of log probabilities (Rafailov et al., 2023).

3 Emulated Fine-Tuning: Decoupling the Scale of Pre-training and Fine-tuning

We now describe the framework of emulated fine-tuning (EFT) and how it enables decoupling the scale of pre-training and fine-tuning, as well as *up-scaling*, a special case of emulated fine-tuning that is particularly useful in practice.

3.1 Preliminaries

Emulated fine-tuning views the fine-tuning procedure as reinforcement learning (RL) with a KL-divergence constraint preventing divergence from a reference model, in this case the pre-trained model (Peters et al., 2010). That is, we view the result of fine-tuning π_{ft} as the solution to

$$\pi_{\text{ft}} = \pi^*(r, \pi_{\text{ref}}) = \arg \max_{\pi} \mathbb{E}_{x \sim p(x), y \sim \pi(\cdot | x)} [r(x, y) - \beta \text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))] \quad (1)$$